Haoyu Ma mahaoyu@umich.edu Website haoyuma

Education Background

University of Michigan at Ann Arbor

M.S. in Robotics
Overall GPA: 4.0/4.0; **08/2024 - Present**

University of North Carolina at Chapel Hill

B.S. in Applied Mathematics, **B.S.** in Physics
Overall GPA: 3.918/4.0;
GPA of Applied Mathematics: 3.89/4.0; GPA of Physics: 3.96/4.0 **08/2020 - 05/2024**

Publication
*denotes equal contribution

Song, J.*, **Ma, H.***, Bagoren, O., Sethuraman, A. V., Zhang, Y., Skinner, K. A., "*OceanSim: A GPU-Accelerated Underwater Robot Perception Simulation Framework,*" *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2025.* **10/2025 Hangzhou, CN**

Ma, H., Tamim, S. I., Guan, J. H., Sáenz, P. J., "*Solid Particles Walking on a Vibrating Interface.*" **Under Prep**

Conference And Workshop Presentations

"*OceanSim: A GPU-Accelerated Underwater Robot Perception Simulation Framework*"
AQ^2 UASIM Workshop, IEEE International Conference on Robotics And Automation (ICRA), **05/2025 Atlanta, GA**

"*Solid Particles Walking on a Vibrating Interface*"
Oral Presentation, APS Division of Fluid Dynamics (DFD), **11/2024 Salt Lake City, UT**

Research Project

Training Underwater Object Detection with Synthetic Data **Ann Arbor, MI**
Supervised by Prof. Katherine Skinner, Field Robotics Group (FRoG) **05/2025 - Present**

➢ Built an object-based synthetic data generation pipeline in Nvidia Omniverse, integrated with realistic real-time underwater rendering effects such as haze and particles for sim2real transfer.
➢ Leveraged SOTA stable-diffusion imageTo3D generative network to replicate real-world assets for solving data scarcity and unbalanced data distribution in real-world datasets with synthetic data.
➢ Training object detection model such as YOLO, Mask R-CNN and segmentation like MaskFormer with synthetic data that increases robustness of perception performance in highly degraded and complex-structured environment.
➢ Extensive applications of OpenUSD for non-destructive and systemic scene authoring and shader coding for domain randomization.

Teleoperating KUKA Arm with Simulation-in-the-loop Digital Twin **Ann Arbor, MI**
Supervised by Prof. Dmitry Berenson, Auto Robotic Manipulation Lab (ARM) **05/2025 - 08/2025**

➢ Developed simulation-in-the-loop tele-operation of Kuka arm in Isaac Sim environment with Apple Vision VR set, ROS2, and MoveIt motion planning.
➢ Leveraged Isaac SDF toolkit for high fidelity simulation of forced manipulation task such as closing oil cap or screwing bolts.
➢ Used PhysX joint simulation for replicating deformable object behaviors such as bending due to gravity and fracture upon intense collision, minimizing sim2real transfer for the trained policy.
➢ The simulation throughput can reach 20Hz+ for both vision and force signal on a RTX 3090 machine, enabling both downstream policy training and later online virtual inference for testing.

Underwater Robots Digital Twin **Ann Arbor, MI**
Supervised by Prof. Katherine Skinner, Field Robotics Group (FRoG) **01/2025 - 05/2025**

➢ 3D Reconstruction of Marine Hydrodynamics Laboratory Towing Tank (MHL, Department of Marine Engineering, UoM) and real-world interactive objects from lidar scan and stereo camera.

➢ Created customizable and high visual fidelity digital twin of common underwater environments and AUV such as BlueROV robot in Isaac Sim, interfaced with ROS2.
➢ Developed GPU-accelerated underwater sensor (Imaging/Side Scan Sonar, DVL, etc) through Nvidia Warp, a powerful package exposes with python API and JIT compiled to CUDA C++ code .
➢ Simulated realistic underwater column effects suitable to any Jerlov water types by combining Underwater Image Formation Model and real-time PBR in Omniverse RTX renderer.

Solid Particles Walking on a Vibrating Interface **Chapel Hill, NC**
Supervised by Pedro Sáenz, Physical Mathematics Lab (PML) **05/2023 - 05/2024**
➢ Discovered period-doubling vertical dynamics and "walking" states for solid micro-particles ∼0.4mm diameter on a vibrated liquid-liquid interface.
➢ Characterized the "walking" regimes and speed by grayscale image processing and classic object tracking with Kalman Filter.
➢ Numerically computed and simulated particles' hydrodynamical behaviors with self-developed PDE numerical solver in MATLAB.
➢ Designed and assembled the primary experiment structure for vibrational fluid dynamic research with the Modal Shop electrodynamic shaker.

Scholarship and Awards

Rackham International Student Fellowship Robotics Department Finalist
➢ Two finalists per department will be selected for College of Engineering for fellowship competition. *Department of Robotics, UMich* **08/2025**

Daniel C. Johnson Outstanding Junior Award (Top 1 within department)
➢ Awarded annually to the physics major who is judged by the faculty to be the most outstanding student of the junior class. *Department of Physics and Astronomy, UNC* **04/2023**

Summer Undergraduate Research Fellowship (54 recipients, 2023)
➢ Granted to undergraduate students conducting research under a faculty mentor. $4000 fellowship included. *Department of Undergraduate Research, UNC* **04/2023**

Skills

Software/Computing Literacy: Isaac Sim/Lab, OpenUSD, Omniverse Kit, ROS2, Python, C/C++, PyTorch, GLSL, CUDA, Docker, SolidWorks, Arduino, LabView, Unreal Engine, Illustrator, Premiere Pro, Adobe Substance Combo

Experimental Techniques: High-Speed Videography, 3D Printing, Laser Cutting, Basic Electronics, Metal and Wood Crafting, NI-DAQ based Signal Processing, Particle Image Velocimetry

Professional Membership: *IEEE* and *APS* (American Physical Society)

Services: Reviewer of *IROS*, *ICRA*

References

Field Robotics Group (FRoG)
Prof. Katherine Skinner
Assistant Professor
Robotics, UoM
kskin@umich.edu

Physical Mathematics Lab (PML)
Prof. Pedro Sáenz
Assistant Professor
Applied Mathematics, UNC
saenz@unc.edu